FRANCESCO BASSIGNANA

FULL STACK & GENERATIVE AI ENGINEER

Cuneo, Italy
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Profile

Full-stack and generative AI engineer with a BSc in Computer Science and MSc studies in Artificial Intelligence Systems. I build end-to-end systems across React and Next.js frontends, TypeScript and Python services, databases, containerized delivery and machine-learning workflows. My research covers LLM evaluation, generative models for synthetic data, computer vision and audio reconstruction. I am strongest where software engineering and applied AI meet: turning experiments into usable services, evaluation platforms and reliable integrations.

Technical Skills

Full Stack: TypeScript, JavaScript, React, Next.js, Node.js, Python, FastAPI, REST APIs

Generative AI and ML: PyTorch, LLMs, Prompt Engineering, LangChain, Ollama, Diffusion Models, Flow Matching, Computer Vision

Data and Delivery: PostgreSQL, MongoDB, Docker, AWS, Git, Model Evaluation, Experiment Design

Systems: Go, C, MQTT, Zigbee, IoT Integration

Experience

RESEARCH INTERN — GENERATIVE AI & COMPUTER VISION

École de technologie supérieure (ÉTS) — Montréal, Canada

2026

Researching synthetic infrared-image generation with diffusion models and flow matching. Training, fine-tuning and comparing generative models, evaluating output quality and investigating how synthetic data can support downstream computer-vision tasks on specialized real-world imagery.

Skills: Python, PyTorch, Generative AI, Diffusion Models, Flow Matching, Model Evaluation

RESEARCH INTERN — LLM EVALUATION & FULL-STACK SYSTEMS

University of Turin

2024 - 2024

Researched system-prompting strategies for LLMs across coding, education, summarization and content creation. Designed and developed a full-stack evaluation platform with Next.js, React, LangChain, Ollama and local language models; organized expert comparisons and analyzed quantitative and qualitative feedback.

Skills: Next.js, React, LangChain, Ollama, LLMs, Prompt Engineering

FULL-STACK SOFTWARE & IOT CONSULTANT

S.H.D. Elettronica S.r.l.

2023 - 2024

Designed and developed web, software and connected-device solutions for operational use cases. Integrated MQTT, Zigbee, HTTP/REST services and embedded components for a port-focused IoT system, contributing across software design, implementation and system integration.

Skills: Full-Stack Development, REST APIs, MQTT, Zigbee, IoT, Firmware

Education

MSc ARTIFICIAL INTELLIGENCE SYSTEMS

University of Trento

2024 - 2026

Advanced study of machine learning and AI engineering, with focus areas in generative models, computer vision, NLP and audio signal processing.

Skills: AI Engineering, Deep Learning, Computer Vision, NLP

BSc COMPUTER SCIENCE

University of Turin

2021 - 2024

Foundation in software engineering, algorithms, databases, operating systems and computer networks.

Skills: Software Engineering, Algorithms, Databases, Networks

Selected Full-Stack and Generative AI Projects

LLM PROMPT EVALUATION PLATFORM

Full Stack / Generative AI

Built a web platform for comparing system-prompt strategies across software engineering and other knowledge-work tasks. Integrated local OpenChat, Mistral and Llama 3 models, structured expert evaluations and analyzed both numerical results and written feedback to derive practical prompting guidelines.

Skills: Next.js, React, LangChain, Ollama, Local LLMs

GitHub: https://github.com/bassi-uni/tesi

DIGITAL PHOTO AI

Full Stack / Computer Vision

Developed a portrait-framing application that detects faces and produces document-photo crops under configurable pixel and physical-dimension constraints. Implemented a React client, FastAPI service, OpenCV image pipeline, YOLO-based face detection and downloadable image and metadata outputs.

Skills: React, FastAPI, Python, OpenCV, YOLO, REST API

GitHub: https://github.com/bacobax/digital_photo_ai

AUDIO GAP FILLING

Generative AI / Audio

Building a configurable research framework for reconstructing missing audio regions from highly limited training data. Implemented reusable YAML-driven training pipelines, masked-autoencoder reconstruction over mel-spectrogram patches, VAE waveform compression and evaluation utilities, with latent-diffusion integration in progress.

Skills: PyTorch, Masked Autoencoders, Vision Transformers, VAE, Diffusion Models

GitHub: https://github.com/bacobax/audio_overfit_impainting

Languages

- Italian — Native
- English — B2, Cambridge certified